

Build an End to End Image Classification/Recognition Application

March 07, 2022

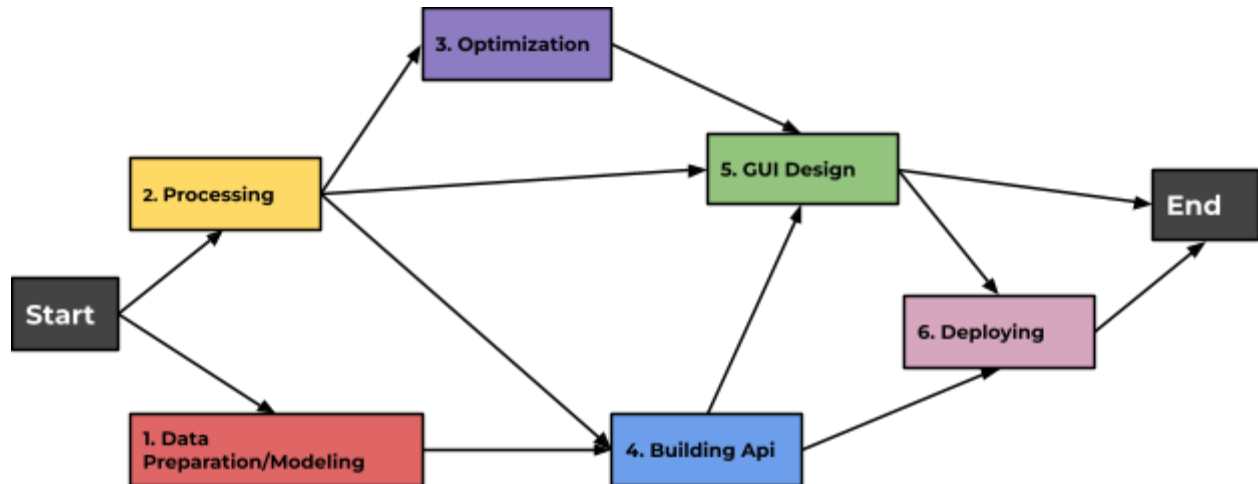
Overview

This project consists in building an image recognition application that enables to classify webcam images using deep learning. This face recognition system could be deployed at entry gates of office buildings to provide a daily report of employees checking in at the office hours.

Goals

1. **Face Recognition:** Using Webcam to recognise employees at the entry gate of office building.
2. **Optimization :** Optimization of face recognition algorithm based on deep learning multi feature fusion driven by big data and Real-time Image Processing using Spark.
3. **Image Classification:** It is used for distinguishing between multiple image sets.
4. **Daily reporting:** Using power BI suite for optimal graphical display.
5. **GUI :** Web application and/or Android app.

Dependency Network Diagram



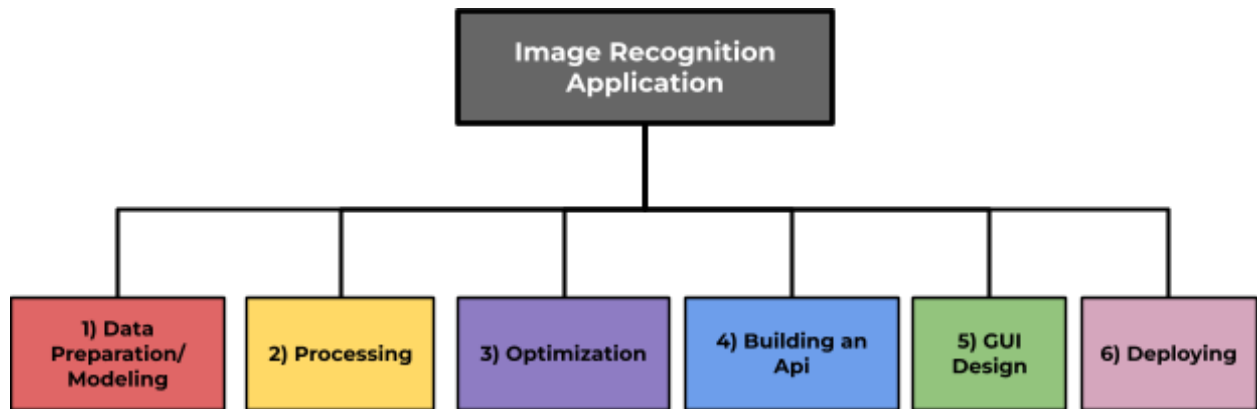
Specifications

Data would be in the form of images i.e. pictures. Pictures are a matrix of pixels. Images would be required in a larger number for building the entire end to end application. The images captured need to be of a high resolution and can be slightly distorted. There can be some amount of noise present in the images so that the algorithm is able to classify the images properly.

Milestones

1. Data preparation and modeling using Deep Neural Networks
2. Parallel video processing on Apache Hadoop clusters.
3. Optimization of face recognition algorithms using multi-fusion features in the big data environment.

4. Building an Api.
5. Designing the user Interface.
6. Integration of the user interface and modeling.



Dependency Structure Matrix (DSM)

ACTIVITY	DESCRIPTION	PREDECESSORS	DURATION WEEKS
1	Data preparation/modeling	-	1
2	Processing	1	5
3	Optimization	1,2	4
4	Building an Api	1,2	2
5	GUI Design	1,2,3,4	3
6	Deploying	1,2,3,4,5	1
TOTAL			3 Months

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						